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Java SE 27 & JDK 27

Interface ObjectStreamConstants

All Known Implementing Classes:

ObjectInputStream, ObjectOutputStream

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public interface **ObjectStreamConstants**

Constants written into the Object Serialization Stream.

Since:

1.1

Field Summary

Fields

Modifier and Type	Field	Description
static final int	baseWireHandle	First wire handle to be assigned.
static final int	PROTOCOL_VERSION_1	A Stream Protocol Version.
static final int	PROTOCOL_VERSION_2	A Stream Protocol Version.
static final byte	SC_BLOCK_DATA	Bit mask for ObjectStreamClass flag.
static final byte	SC_ENUM	Bit mask for ObjectStreamClass flag.
static final byte	SC_EXTERNALIZABLE	Bit mask for ObjectStreamClass flag.
static final byte	SC_SERIALIZABLE	Bit mask for ObjectStreamClass flag.
static final byte	SC_WRITE_METHOD	Bit mask for ObjectStreamClass flag.
static final SerializablePermissi	SERIAL_FILTER_PERMISSION	Enable setting the system-wide serial filter.
static final short	STREAM_MAGIC	Magic number that is written to the stream header.

static final short	**STREAM_VERSION**	Version number that is written to the stream header.
static final SerializablePermissi	**SUBCLASS_IMPLEMENTATION_**	Enable overriding of readObject and writeObject.
static final SerializablePermissi	**SUBSTITUTION_PERMISSION**	Enable substitution of one object for another during serialization/ deserialization.
static final byte	**TC_ARRAY**	new Array.
static final byte	**TC_BASE**	First tag value.
static final byte	**TC_BLOCKDATA**	Block of optional data.
static final byte	**TC_BLOCKDATALONG**	long Block data.
static final byte	**TC_CLASS**	Reference to Class.
static final byte	**TC_CLASSDESC**	new Class Descriptor.
static final byte	**TC_ENDBLOCKDATA**	End of optional block data blocks for an object.
static final byte	**TC_ENUM**	new Enum constant.
static final byte	**TC_EXCEPTION**	Exception during write.
static final byte	**TC_LONGSTRING**	Long string.
static final byte	**TC_MAX**	Last tag value.
static final byte	**TC_NULL**	Null object reference.
static final byte	**TC_OBJECT**	new Object.
static final byte	**TC_PROXYCLASSDESC**	new Proxy Class Descriptor.
static final byte	**TC_REFERENCE**	Reference to an object already written into the stream.
static final byte	**TC_RESET**	Reset stream context.

static final byte	TC_STRING	new String.

Field Details

STREAM_MAGIC

```
static final short STREAM_MAGIC
```

Magic number that is written to the stream header.

See Also:

Constant Field Values

STREAM_VERSION

```
static final short STREAM_VERSION
```

Version number that is written to the stream header.

See Also:

Constant Field Values

TC_BASE

```
static final byte TC_BASE
```

First tag value.

See Also:

Constant Field Values

TC_NULL

```
static final byte TC_NULL
```

Null object reference.

See Also:

Constant Field Values

TC_REFERENCE

```
static final byte TC_REFERENCE
```

Reference to an object already written into the stream.

See Also:

Constant Field Values

TC_CLASSDESC

`static final byte TC_CLASSDESC`

new Class Descriptor.

See Also:

Constant Field Values

TC_OBJECT

`static final byte TC_OBJECT`

new Object.

See Also:

Constant Field Values

TC_STRING

`static final byte TC_STRING`

new String.

See Also:

Constant Field Values

TC_ARRAY

`static final byte TC_ARRAY`

new Array.

See Also:

Constant Field Values

TC_CLASS

`static final byte TC_CLASS`

Reference to Class.

See Also:

Constant Field Values

TC_BLOCKDATA

```
static final byte TC_BLOCKDATA
```

Block of optional data. Byte following tag indicates number of bytes in this block data.

See Also:

Constant Field Values

TC_ENDBLOCKDATA

```
static final byte TC_ENDBLOCKDATA
```

End of optional block data blocks for an object.

See Also:

Constant Field Values

TC_RESET

```
static final byte TC_RESET
```

Reset stream context. All handles written into stream are reset.

See Also:

Constant Field Values

TC_BLOCKDATALONG

```
static final byte TC_BLOCKDATALONG
```

long Block data. The long following the tag indicates the number of bytes in this block data.

See Also:

Constant Field Values

TC_EXCEPTION

```
static final byte TC_EXCEPTION
```

Exception during write.

See Also:

Constant Field Values

TC_LONGSTRING

```
static final byte TC_LONGSTRING
```

Long string.

See Also:

Constant Field Values

TC_PROXYCLASSDESC

`static final byte TC_PROXYCLASSDESC`

new Proxy Class Descriptor.

See Also:

Constant Field Values

TC_ENUM

`static final byte TC_ENUM`

new Enum constant.

Since:

1.5

See Also:

Constant Field Values

TC_MAX

`static final byte TC_MAX`

Last tag value.

See Also:

Constant Field Values

baseWireHandle

`static final int baseWireHandle`

First wire handle to be assigned.

See Also:

Constant Field Values

SC_WRITE_METHOD

`static final byte SC_WRITE_METHOD`

Bit mask for ObjectStreamClass flag. Indicates a Serializable class defines its own writeObject method.

See Also:

Constant Field Values

SC_BLOCK_DATA

```
static final byte SC_BLOCK_DATA
```

Bit mask for ObjectStreamClass flag. Indicates Externalizable data written in Block Data mode. Added for PROTOCOL_VERSION_2.

Since:

1.2

See Also:

PROTOCOL_VERSION_2, Constant Field Values

SC_SERIALIZABLE

```
static final byte SC_SERIALIZABLE
```

Bit mask for ObjectStreamClass flag. Indicates class is Serializable.

See Also:

Constant Field Values

SC_EXTERNALIZABLE

```
static final byte SC_EXTERNALIZABLE
```

Bit mask for ObjectStreamClass flag. Indicates class is Externalizable.

See Also:

Constant Field Values

SC_ENUM

```
static final byte SC_ENUM
```

Bit mask for ObjectStreamClass flag. Indicates class is an enum type.

Since:

1.5

See Also:

Constant Field Values

SUBSTITUTION_PERMISSION

```
static final SerializablePermission SUBSTITUTION_PERMISSION
```

Enable substitution of one object for another during serialization/ deserialization.

Since:

1.2

See Also:

`ObjectOutputStream.enableReplaceObject(boolean)`, `ObjectInputStream.enableResolveObject(boolean)`

SUBCLASS_IMPLEMENTATION_PERMISSION

```
static
final SerializablePermission SUBCLASS_IMPLEMENTATION_PERMISSION
```

Enable overriding of readObject and writeObject.

Since:

1.2

See Also:

`ObjectOutputStream.writeObjectOverride(Object)`, `ObjectInputStream.readObjectOverride()`

SERIAL_FILTER_PERMISSION

```
static final SerializablePermission SERIAL_FILTER_PERMISSION
```

Enable setting the system-wide serial filter.

Since:

9

See Also:

`ObjectInputFilter.Config.setSerialFilter(ObjectInputFilter)`

PROTOCOL_VERSION_1

```
static final int PROTOCOL_VERSION_1
```

A Stream Protocol Version.

All externalizable data is written in JDK 1.1 external data format after calling this method. This version is needed to write streams containing Externalizable data that can be read by pre-JDK 1.1.6 JVMs.

Since:

1.2

See Also:

`ObjectOutputStream.useProtocolVersion(int)`,
Constant Field Values

PROTOCOL_VERSION_2

```
static final int PROTOCOL_VERSION_2
```

A Stream Protocol Version.

This protocol is written by JVM 1.2.

Externalizable data is written in block data mode and is terminated with TC_ENDBLOCKDATA. Externalizable class descriptor flags has SC_BLOCK_DATA enabled. JVM 1.1.6 and greater can read this format change.

Enables writing a nonSerializable class descriptor into the stream. The serialVersionUID of a nonSerializable class is set to 0L.

Since:

1.2

See Also:

`ObjectOutputStream.useProtocolVersion(int)`,
`SC_BLOCK_DATA`,
Constant Field Values

Report a bug or suggest an enhancement
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